

SYDNEY TECHNICAL HIGH SCHOOL

YEAR 12 ASSESSMENT TASK 3

JUNE 2004

MATHEMATICS: 2 UNIT

Time Allowed: 70 minutes

Instructions:

- Write your name and class at the top of this page, and at the top of each answer sheet
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions are of equal value. Attempt all questions.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Standard integrals are attached at the back of this paper.

Name:

Teacher:

Q1	Q2	Q3	Q4	Q5	Total
/12	/12	/12	/12	/12	/60

Question 1 (12 Marks)

a) For the parabola $x^2 - 4x - 8y - 36 = 0$

find : i) the co-ordinates of the vertex (1)

ii) the focal length (1)

iii) the co-ordinates of the focus (1)

iv) the equation of the directrix (1)

b) If $x^{1.0986} = 3$ find:

i) $\log_x 3$ (1)

ii) $\log_x 81$ (1)

c) Evaluate $\int_0^{\pi/2} \sin 2x \, dx$ (2)

d) Differentiate $y = x \tan x$ (2)

e) Differentiate $y = \log_e(4 - x^2)$ (2)

Question 2

a) Sketch the graph of $y = \cos \frac{x}{4}$ for $-4\pi \leq x \leq 4\pi$ (2)

b) (i) Evaluate $\int_0^{2\pi} \cos \frac{x}{4} dx$ (2)

(ii) For the function in (i) $y = \cos \frac{x}{4}$, complete the table below

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
y					

Hence approximate $\int_0^{2\pi} \cos \frac{x}{4} dx$ using Simpson's rule with 5 function values and answering correct to 2 decimal places. (3)

Question 2 (cont)

(iii) Find the percentage error given by using Simpson's rule to approximate the integral (answer correct to 1 decimal place). (1)

- c) (i) Find the co-ordinates of the point on the curve $y = x \ln x$ for which the gradient of the tangent is equal to 2. (2)
- (ii) Hence find the equation of the normal at this point. (2)

Question 3

a) Solve the equation $5e^{2x} = 0.4213$. Give your answer to 4 decimal places. (2)

b) By using the substitution $u = e^x$ or otherwise, solve the equation $e^{2x} - 6e^x + 5 = 0$
Answer to 3 decimal places where necessary (3)

c) Solve for θ in the domain $0 \leq \theta \leq 2\pi$: (3)

$$4 \sin \theta - 3 \cos \theta = 0$$

d) Differentiate (2)

$$y = \sqrt{\sin 2x}$$

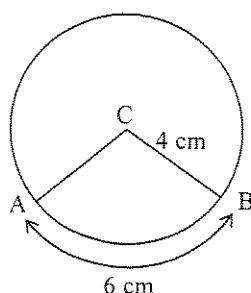
e) Find the derivative of $y = e^{\cos x}$ (1)

Question 4

- a) Find
- i) $\int e^{5x} dx$ (1)
- ii) $\int \sqrt{e^x} dx$ (2)
- b) Express $\frac{2\pi^c}{5}$ in degrees (1)

Question 4 (cont.)

c)



In the diagram, AB is an arc of length 6cm in a circle, centre C, radius 4cm. Find the

- i) size of angle ACB in radians (1)
- ii) area of sector ACB (1)
- iii) area of the minor segment cut off by the chord AB (2)

- d) (i) Find the area bounded by the x axis, the curve $y = e^x$ and the lines

$$x = 0 \text{ \& } x = \log_e 5. \quad (2)$$

- (ii) This area is rotated about the x axis.

Find the volume of the solid so formed. (Leave answer in terms of π). (2)

Question 5

a) Solve for x , $\sin^2 2x = \frac{1}{4}$ ($0 \leq x \leq 180^\circ$) (2)

- b) (i) Find the value of k for which the equation $x^2 - (k-5)x + (k-7) = 0$ has roots which are reciprocals of one another. (1)

- (ii) If α and β are the roots of this equation find

A) $\alpha + \beta$ (1)

B) $\frac{1}{\alpha} + \frac{1}{\beta}$ (1)

c) Find the values of P, Q and R if $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$. (3)

d) Find all the values of k for which $(k+2)x^2 + x - 3 = 0$ has real roots. (2)

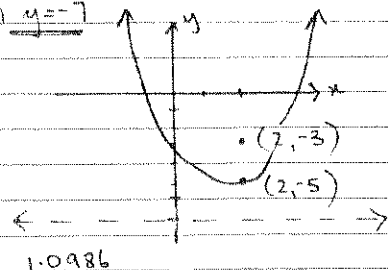
e) Evaluate $\int_1^2 \frac{x^3 - 3x + 2}{x^2} dx$ (2)

End of Exam.

Question 1

a) $x^2 - 4x + 4 = 8y + 36 + 4$
 $(x-2)^2 = 8y + 40$
 $(x-2)^2 = 8(y+5)$

- i) Vertex (2, -5) ①
 ii) $4a = 8 \therefore a = 2$ ①
 iii) Focus (2, -3) ①
 iv) $y = -7$ ①



b) $x = 3$
 i) $\log_3 3 = 1.0986$ ①
 ii) $\log_8 1 = \log_3 3$
 $= 4 \log_3 3$
 $= 4.3944$ ①

c) $\int_0^{\pi/3} \sin 2x \, dx = \left[-\frac{1}{2} \cos 2x \right]_0^{\pi/3}$
 $= -\frac{1}{2} \left[\cos 2\pi/3 - \cos 0 \right]$
 $= -\frac{1}{2} \left[-\frac{1}{2} - 1 \right]$
 $= 3/4$ ①

d) $\frac{d}{dx} (x \tan x)$

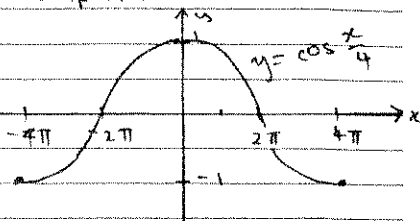
$u = x \quad v = \tan x$
 $u' = 1 \quad v' = \sec^2 x$
 $\frac{dy}{dx} = \tan x + x \sec^2 x$ ②

e) $\frac{d}{dx} \ln(4-x^2) = \frac{-2x}{4-x^2}$ ②

Question 2

a) period = $\frac{2\pi}{1/4} = 8\pi$

amplitude = 1



b) i) $\int_0^{2\pi} \cos \frac{x}{4} \, dx$
 $= \left[4 \sin \frac{x}{4} \right]_0^{2\pi}$
 $= 4 \left(\sin \frac{\pi}{2} - \sin 0 \right)$
 $= 4$

	F	L
x	0	$\pi/2$
y	1	0

ii) $S = \frac{\pi/2}{3} \left[1 + 0 + 4(0.924 + 0.383) + 2 \times 0.707 \right]$
 $= 4.00$

c) i) $y = x \ln x \therefore \frac{dy}{dx} = 2$

$u = x \quad v = \ln x$
 $u' = 1 \quad v' = \frac{1}{x}$
 $\frac{dy}{dx} = \ln x + 1$

$\ln x + 1 = 2$
 $\ln x = 1$

$\log_e x = 1 \therefore e = x$

$\therefore$ pt (e, e)

ii) $m = -\frac{1}{2}$ at (e, e)

eqn $y - e = -\frac{1}{2}(x - e)$

$2y - 2e = -x + e$

$x + 2y - 3e = 0$

Question 3

a) $5e^{2x} = 1.4213$
 $e^{2x} = \frac{1.4213}{5}$

$2x \ln e = \ln(0.28426)$
 $x = -1.2369$ (4 d.p.)

b) $e^{2x} - 6e^x + 5 = 0$
 $u^2 - 6u + 5 = 0$
 $(u - 5)(u - 1) = 0$
 $u = 5 \quad u = 1$
 $\therefore x = 1.609 \quad x = 0$

c) $4 \sin \theta - 3 \cos \theta = 0$
 $4 \sin \theta = 3 \cos \theta$
 $\tan \theta = \frac{3}{4}$
 $\theta = 0.64^\circ, 3.79^\circ$

d) $y = (\sin 2x)^{1/2}$
 $\frac{dy}{dx} = \frac{1}{2} (2 \cos 2x) (\sin 2x)^{-1/2}$
 $= \frac{\cos 2x}{\sqrt{\sin 2x}}$

e) $\frac{d}{dx} (e^{\cos x}) = -\sin x \cdot e^{\cos x}$

Question 4

a) i) $\int e^{5x} \, dx = \frac{1}{5} e^{5x} + C$

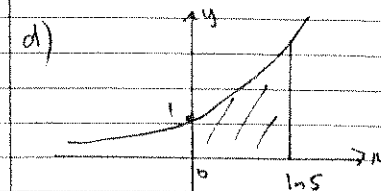
ii) $\int e^{x/2} \, dx = 2e^{x/2} + C$
 OR $2\sqrt{e^x} + C$

b) $\frac{2\pi}{5} = 72^\circ$

c) i) $b = 4 \times \theta$
 $\theta = 1.5^\circ$
 $\therefore \angle ACB = 1.5^\circ$

ii) $A = \frac{1}{2} \times 4^2 \times 1.5$
 $= 12 \text{ cm}^2$

iii) $A = \frac{1}{2} \times 4^2 (1.5 - \sin 1.5)$
 $= 4.02 \text{ cm}^2$

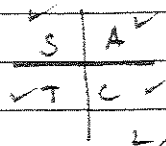


d) i) $A = \int_1^5 e^x \, dx$
 $= \left[e^x \right]_1^5$
 $= e^5 - e^1$
 $= 4e^2$

ii) $V = \pi \int_0^{\ln 5} e^{2x} \, dx$
 $= \pi \left[\frac{e^{2x}}{2} \right]_0^{\ln 5}$
 $= \frac{\pi}{2} [e^{2 \ln 5} - e^0]$
 $= \frac{\pi}{2} [25 - 1]$
 $= 12\pi \text{ unit}^3$

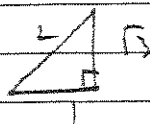
Question 5

a) $\sin^2 2x = \frac{1}{4}$



$\sin 2x = \pm \frac{1}{2}$

acute $2x = 30^\circ$



$\therefore 2x = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

$x = 15^\circ, 75^\circ, 105^\circ, 165^\circ$

b) i) Reciprocal roots $\alpha, \frac{1}{\alpha}$

$\therefore$ product is 1 $\therefore \frac{c}{a} = 1$

$k-7 = 1$

1

$k = 8$

ii) A) $\alpha + \beta = \frac{k-5}{1} = \underline{3}$

B) $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \underline{\underline{\frac{3}{1}}}$

c) $3x^2 + 5x - 1 = P(x+1)^2 + Q(x+1) + R$
 $= P(x^2 + 2x + 1) + Qx + Q + R$
 $= Px^2 + 2Px + P + Qx + Q + R$

$= P(x^2) + x(2P+Q) + P+Q+R$

$\therefore P = 3 \quad 2P+Q = 5 \quad 3-1+R = -1$

$6+Q = 5$

$R = -3$

$Q = -1$

d) Real Roots $\Delta \geq 0 \quad 1 - 4(k+2)x - 3 \geq 0$

$1 + 12(k+2) \geq 0$

$1 + 12k + 24 \geq 0$

$12k \geq -25$

$k \geq \frac{-25}{12}$

e) $\int_1^2 \frac{3x^2 - 3x + 2}{x^2} dx = \int_1^2 x - \frac{3}{x} + 2x^{-2} dx$

$= \left[\frac{x^2}{2} - 3 \ln x - \frac{2}{x} \right]_1^2$

$= \left[(2 - 3 \ln 2 - 1) - \left(\frac{1}{2} - 0 - 2 \right) \right] = \underline{\underline{2\frac{1}{2} - 3 \ln 2}}$